IBM SKILLSBUILD UNIVERSITY EDUCATION

OFFICIAL PROJECT SUBMISSION REPORT — EDUTRANSLATE AI

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TRACK DOMAIN: Education

PROBLEM STATEMENT ID: Problem Statement No.17 – Multi-language Course Content Translator Agent

TARGET ARCHITECTURE: Pure-Python Local RAG Execution Pipeline

FOUNDATION LLM ENGINE: ibm/granite-4-h-small (IBM watsonx.ai Platform)

DEPLOYMENT REGION: Dallas (us-south Endpoint)

1. PROBLEM STATEMENT & CRITICAL HIGHER EDUCATION CHALLENGES

Academic course frameworks, complex scientific lecture text files, and advanced syllabus sheets are traditionally locked behind severe language barriers for diverse student populations. Non-native students struggle to access high-quality engineering, computational, and technical reference materials because standard machine translation tools fail under the following engineering limitations:

* Out-of-Memory Output Truncation: Long multi-page documents frequently exceed base context windows, causing models to drop over 50% of the content or arbitrarily switch to a short summary instead of a literal translation.
* Structural Layout Loss: Complex page boundaries, spacing headers, and document alignments are often flattened or broken during standard text parsing loops.
* Technical Terminology Drift: Literal translation engines distort domain-specific pedagogical contexts, substituting verified regional curriculum keywords with incorrect literal phrases.

2. PROPOSED SOLUTION ARCHITECTURE & OPERATIONAL ROUTINE

EduTranslate AI is a production-grade multi-language educational translation agent built programmatically using the enterprise 'ibm_watsonx_ai' Python SDK. The system completely bypasses browser-based playgrounds to establish a direct HTTPS client link to the advanced 'ibm/granite-4-h-small' foundation model instance running on the IBM Cloud Dallas datacenter layer.

The application architecture introduces an end-to-end automated processing pipeline:

* Multi-Format Document Parsing: Uses PyMuPDF (fitz), python-docx, and python-pptx modules to extract text page-by-page while preserving original asset structures.
* Streaming Window Chunking: Automatically segments raw text extractions into precise, sequential 1,200-character windows. This mitigation strategy processes documents block-by-block to guarantee 100% full-text translation coverage.
* Local In-Memory Retrieval-Augmented Generation (RAG): A lightweight mathematical vector inference framework built entirely inside native Python. It computes text semantic frequency weights and cosine similarities locally in memory to fetch verified curriculum benchmarks and academic glossaries.
* Prompt Grounding Injector: The retrieved glossary references are dynamically injected into the Granite system prompt context, forcing the language model to maintain absolute technical integrity and instructional accuracy.

3. TECHNICAL NOVELTY & PLATFORM DIFFERENTIATION

* Zero-Dependency Stable RAG Core: Standard Python vector database abstractions (such as default ChromaDB or FAISS) rely heavily on compiled background binary engines written in C++ or Rust (like 'hnswlib'). On Windows development setups, these routinely trigger severe system compilation and wheel failures unless massive Microsoft Visual C++ Build Tools are manually installed. EduTranslate AI eliminates these vulnerabilities entirely by executing vector matching via pure-Python standard libraries ('math', 're'). This guarantees zero installation errors on any machine out-of-the-box.

* Bi-Directional Linguistic Vector Swapping: Employs an interactive Streamlit UI navigation row that allows students to flip translation paths instantly (e.g., English ⇄ Hindi) via a single toggle button while preserving workspace session history across active updates.

* Strict Pedagogical Alignment Guardrails: Suppresses chat engine leaks, model hallucinations, and conversational filler, returning exclusively the raw, localized educational output data.